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(54) **CRYSTALS OF 3-HYDROXY-N'-(1,3-DIMETHYLBUTYLIDENE)-2-NAPHTHOIC ACID HYDRAZIDE AND PRODUCTION METHOD FOR THE SAME**

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(2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,380,288 B1 \* 4/2002 Hojo ..... B60C 15/06  
524/495

2003/0175607 A1 9/2003 Isoda et al.  
2020/0263001 A1 8/2020 Yoshizawa et al.

FOREIGN PATENT DOCUMENTS

|    |               |         |                 |
|----|---------------|---------|-----------------|
| EP | 3 805 306     | 4/2021  |                 |
| JP | 9-124840      | 5/1997  |                 |
| JP | 10-330549     | 12/1998 |                 |
| JP | 10330549 A *  | 12/1998 | ..... B60C 1/00 |
| JP | H10330549 A * | 12/1998 |                 |
| JP | 11-292834     | 10/1999 |                 |
| JP | 11292834 A *  | 10/1999 |                 |
| JP | 2000-190704   | 7/2000  |                 |
| JP | 2003-183510   | 7/2003  |                 |
| JP | 2019-6845     | 1/2019  |                 |
| WO | 2017/104467   | 6/2017  |                 |
| WO | 2019/235526   | 12/2019 |                 |

OTHER PUBLICATIONS

English Translation of JPH10330549A (Year: 1998).\*

English Translation JP H11292834 A (Year: 1999).\*

International Search Report (ISR) issued Feb. 9, 2021 in International (PCT) Application No. PCT/JP2020/047329.

Office Action issued Sep. 24, 2024 in Japanese Patent Application No. 2021-565667, with English-language translation.

Maeda, Morikazu, "Designing Formulation (2), Types and properties of compounding agents", The Society of Rubber Industry, Japan, 1978, vol. 51, No. 10, pp. 808-812, with partial English-language translation.

International Preliminary Report on Patentability and Written Opinion of the International Searching Authority issued May 17, 2022 in International (PCT) Application No. PCT/JP2020/047329.

\* cited by examiner

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(57) **ABSTRACT**

Crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising a peak of maximum intensity at an angle of diffraction (2θ) of 6.9±0.2°, 11.4±0.2° and/or 25.2±0.2°. The crystals are useful as rubber additives in tires.

**16 Claims, 1 Drawing Sheet**

FIG. 1

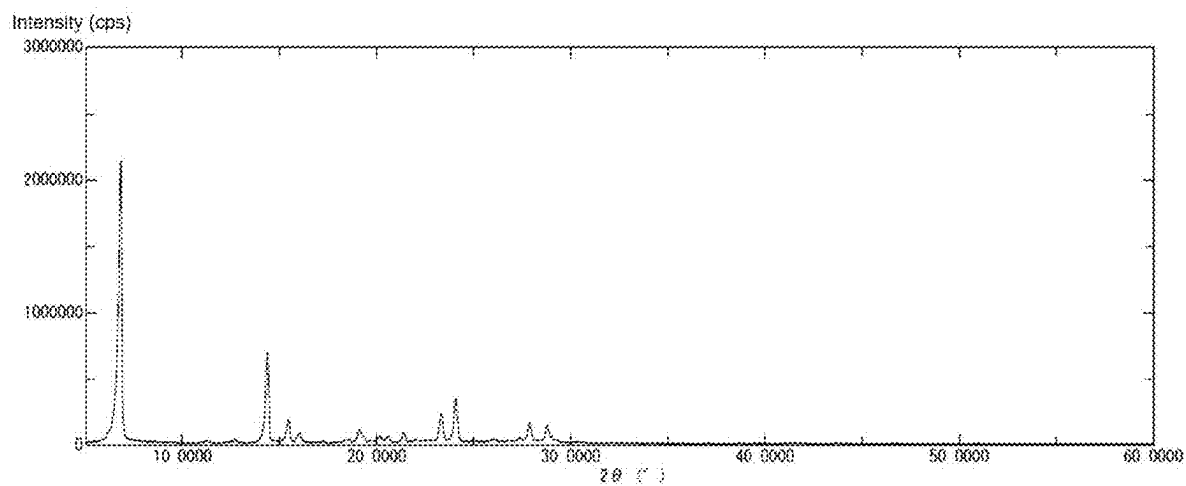
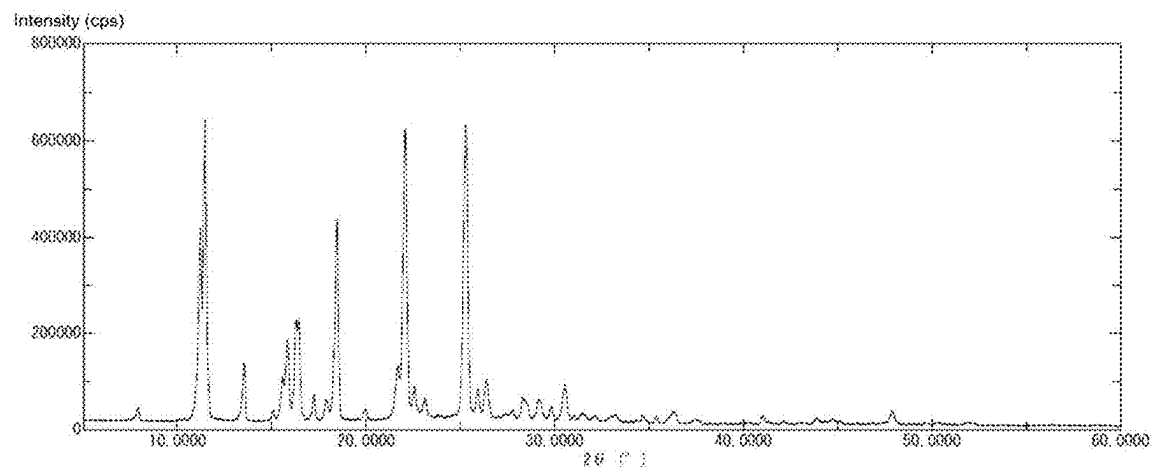


FIG. 2



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**CRYSTALS OF 3-HYDROXY-N'-(1,3-DIMETHYLBUTYLIDENE)-2-NAPHTHOIC ACID HYDRAZIDE AND PRODUCTION METHOD FOR THE SAME**

TECHNICAL FIELD

The present invention relates to novel crystals etc.

BACKGROUND ART

Rubber products, such as tires, contain various additives. For example, a hydrazide compound is used as a rubber additive (for example, see Patent literature 1).

CITATION LIST

Patent Literature

Patent literature 1: JP 2019-6845 A

SUMMARY OF INVENTION

Technical Problem

An object of the present invention is to provide novel crystals.

Another object of the present invention is to provide a novel method for producing the crystals.

Another object of the present invention is to provide a novel rubber additive.

Solution to Problem

The inventors conducted extensive studies to solve the above problems focusing on 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide, and found that novel crystals of the compound can be obtained by appropriately selecting the conditions for precipitation of the crystals (i.e., crystallization).

The inventors also found that the novel crystals include various types of crystals that impart different properties to a rubber component when used as rubber additives.

The inventors further found that first crystals of the present invention have improved low heat build-up as compared with second crystals of the present invention, whereas the second crystals have improved processability as compared with the first crystals. The inventors made further studies and completed the present invention.

That is, the present invention relates to the following.

- (1) Crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising a peak of maximum intensity at an angle of diffraction ( $2\theta$ ) of  $6.9\pm 0.2^\circ$ .
- (2) Crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising characteristic peaks at least at angles of diffraction ( $2\theta$ ) of  $6.9\pm 0.2^\circ$ ,  $14.5\pm 0.2^\circ$ ,  $15.5\pm 0.2^\circ$ ,  $19.1\pm 0.2^\circ$ ,  $23.4\pm 0.2^\circ$ ,  $24.1\pm 0.2^\circ$ ,  $27.9\pm 0.2^\circ$  and  $28.8\pm 0.2^\circ$ .
- (3) Crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising a peak of maximum intensity at an angle of diffraction ( $2\theta$ ) of  $11.4\pm 0.2^\circ$  and/or  $25.2\pm 0.2^\circ$ .
- (4) Crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder

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diffraction pattern comprising characteristic peaks at least at angles of diffraction ( $2\theta$ ) of  $11.2\pm 0.2^\circ$ ,  $11.4\pm 0.2^\circ$ ,  $13.5\pm 0.2^\circ$ ,  $15.5\pm 0.2^\circ$ ,  $15.8\pm 0.2^\circ$ ,  $16.3\pm 0.2^\circ$ ,  $18.4\pm 0.2^\circ$ ,  $21.6\pm 0.2^\circ$ ,  $22.0\pm 0.2^\circ$ ,  $25.2\pm 0.2^\circ$  and  $26.3\pm 0.2^\circ$ .

(5) Crystals comprising the crystals according to the above (1) or (2) and the crystals according to the above (3) or (4).

(6) The crystals according to the above (5), wherein a ratio of the intensity of the peak at an angle of diffraction ( $2\theta$ ) of  $11.4\pm 0.2^\circ$  to that of the peak at an angle of diffraction ( $2\theta$ ) of  $6.9\pm 0.2^\circ$  is 1:99 to 10:90.

(7) The crystals according to any one of the above (1) to (6), wherein the crystals have a melting point of  $151^\circ$  C. or more.

(8) A rubber additive comprising the crystals according to any one of the above (1) to (7).

(9) A rubber composition comprising the rubber additive according to the above (8).

(10) A tire comprising the rubber composition according to the above (9).

(11) A method for producing the crystals according to the above (1) or (2), the method comprising at least forming a crystal nucleus in a solution containing 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide at  $50^\circ$  C. or more.

(12) A method for producing the crystals according to the above (3) or (4), the method comprising at least precipitating crystals from a solution of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide in a solvent containing methanol and methyl isobutyl ketone.

(13) A method for improving low heat build-up and/or processability of a rubber, the method comprising using the crystals according to any one of the above (1) to (7) to improve low heat build-up and/or processability of a rubber.

(14) Use of the crystals according to any one of the above (1) to (7) for improvement of low heat build-up and/or processability of a rubber.

Advantageous Effects of Invention

The present invention provides novel crystals, in particular, novel crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide.

The present invention also provides a novel rubber additive comprising the crystals.

The rubber additive is capable of imparting low heat build-up properties to a rubber component. Thus, a rubber composition comprising the rubber additive can be used to produce a rubber product (such as a tire) with low heat build-up.

In general, a rubber component with low heat build-up tends to have poor processability. However, the rubber additive according to the invention achieves both low heat build-up and good processability.

The present invention also provides a novel method for producing the crystals.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 shows the results of X-ray powder diffraction analysis in Example 1.

FIG. 2 shows the results of X-ray powder diffraction analysis in Example 2.

## DESCRIPTION OF EMBODIMENTS

## Crystals

## Crystals A

The first crystals (crystals A) of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide of the present invention will be described below.

The crystals A may have an X-ray powder diffraction pattern comprising a peak of maximum intensity at an angle of diffraction ( $2\theta$ ) of, for example,  $6.9^\circ$  as analyzed with a powder X-ray diffractometer using Cu-K $\alpha$  rays (wavelength  $\lambda=1.54$  angstroms).

The crystals A may also have an X-ray powder diffraction pattern comprising characteristic peaks at least at angles of diffraction ( $2\theta$ ) of  $6.9^\circ$ ,  $14.5^\circ$ ,  $15.5^\circ$ ,  $19.1^\circ$ ,  $23.4^\circ$ ,  $24.1^\circ$ ,  $27.9^\circ$  and  $28.8^\circ$  as analyzed with a powder X-ray diffractometer using Cu-K $\alpha$  rays (wavelength  $\lambda=1.54$  angstroms).

The crystals A may also have an X-ray powder diffraction pattern comprising characteristic peaks at angles of diffraction ( $2\theta$ ) of, for example,  $6.9^\circ$ ,  $14.5^\circ$ ,  $15.5^\circ$ ,  $16.1^\circ$ ,  $19.1^\circ$ ,  $23.4^\circ$ ,  $24.1^\circ$ ,  $27.9^\circ$  and  $28.8^\circ$  as analyzed with a powder X-ray diffractometer using Cu-K $\alpha$  rays (wavelength  $\lambda=1.54$  angstroms).

In general, an angle of diffraction ( $2\theta$ ) measured by X-ray powder diffractometry may vary to some extent depending on the measurement conditions. Thus the crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide according to the invention include crystals having peaks at angles of diffraction corresponding to the angles of diffraction described above within the margin of error as measured by X-ray powder diffractometry. This error in the measurement of angles of diffraction may be, for example,  $\pm 0.2^\circ$ . That is, the crystals of the present invention include crystals having peaks at angles of diffraction corresponding to the angles of diffraction described above with an error of  $\pm 0.2^\circ$ .

The "angle of diffraction ( $2\theta$ )  $\pm 0.2^\circ$ " herein indicates ( $2\theta - 0.2^\circ$ )  $\leq 2\theta \leq$  ( $2\theta + 0.2^\circ$ ). That is, it means that the angle of diffraction ranges from ( $2\theta - 0.2^\circ$ ) to ( $2\theta + 0.2^\circ$ ).

When the region of the diffraction angle of a peak overlaps with the region of the diffraction angle of another peak, these peaks may overlap with each other. For example, a peak at  $2\theta$  of  $17.7^\circ \pm 0.2^\circ$  overlaps with a peak at  $2\theta$  of  $17.9^\circ \pm 0.2^\circ$  in the angle region from  $17.7^\circ$  to  $17.9^\circ$ , and two distinct peaks may be present in the angle region from  $17.7^\circ$  to  $17.9^\circ$ , or two peaks may overlap with each other in the region from  $17.7^\circ$  to  $17.9^\circ$ .

The crystals A may have a peak with an intensity of, for example, 22 to 75%, preferably 34 to 69%, at an angle of diffraction ( $2\theta$ ) of  $14.5^\circ$  relative to the intensity of the peak at an angle of diffraction ( $2\theta$ ) of  $6.9^\circ$  taken as 100%.

The peaks may have a different intensity from those described above.

The melting point or endothermic peak of the crystals A is  $151^\circ$  C. or more, preferably  $155$  to  $159^\circ$  C., and further preferably  $156$  to  $158^\circ$  C., as determined by differential scanning calorimetry (DSC).

The crystals A may be in any form, and may be in the form of aggregates or a pulverized powder obtained by appropriate means.

The mean particle size of the crystals A determined as a particle size at 50% ( $D_{50}$ ) in the cumulative distribution of the number of particles may be, for example,  $2 \mu\text{m}$  or more, and is preferably  $5$  to  $90 \mu\text{m}$ , and more preferably  $6$  to  $20 \mu\text{m}$ , but not limited thereto.

When the crystals are in the form of aggregates, the mean particle size of the crystals may be appropriately adjusted to the above range by means of, for example, pulverization.

The mean particle size ( $D_{50}$ ) herein may be determined by any method, including the method described in Examples below.

The specific surface area of the crystals A may be, for example,  $0.2$  to  $5 \text{ m}^2/\text{g}$ , and is preferably  $0.2$  to  $3 \text{ m}^2/\text{g}$ , and more preferably  $0.4$  to  $2 \text{ m}^2/\text{g}$ , but not limited thereto.

The specific surface area herein may be determined by any method, including the method described in Examples below.

The crystals A may have any shape, including a plate shape, a pillar shape and an acicular shape, and may preferably have a pillar shape.

## Crystals B

The second crystals (crystals B) of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide of the present invention will be described below.

The crystals B may have an X-ray powder diffraction pattern comprising a peak of maximum intensity at an angle of diffraction ( $2\theta$ ) of, for example,  $11.4^\circ$  and/or  $25.2^\circ$  as analyzed with a powder X-ray diffractometer using Cu-K $\alpha$  rays (wavelength  $\lambda=1.54$  angstroms).

The crystals B may also have an X-ray powder diffraction pattern comprising characteristic peaks at least at angles of diffraction ( $2\theta$ ) of  $11.2^\circ$ ,  $11.4^\circ$ ,  $13.5^\circ$ ,  $15.5^\circ$ ,  $15.8^\circ$ ,  $16.3^\circ$ ,  $18.4^\circ$ ,  $21.6^\circ$ ,  $22.0^\circ$ ,  $25.2^\circ$  and  $26.3^\circ$  as analyzed with a powder X-ray diffractometer using Cu-K $\alpha$  rays (wavelength  $\lambda=1.54$  angstroms).

The crystals B may also have an X-ray powder diffraction pattern comprising characteristic peaks at angles of diffraction ( $2\theta$ ) of, for example,  $11.2^\circ$ ,  $11.4^\circ$ ,  $13.5^\circ$ ,  $15.5^\circ$ ,  $15.8^\circ$ ,  $16.3^\circ$ ,  $17.2^\circ$ ,  $18.4^\circ$ ,  $21.6^\circ$ ,  $22.0^\circ$ ,  $25.2^\circ$ ,  $26.3^\circ$  and  $29.1^\circ$  as analyzed with a powder X-ray diffractometer using Cu-K $\alpha$  rays (wavelength  $\lambda=1.54$  angstroms).

The crystals B may have a peak with an intensity of, for example, 10 to 87% at an angle of diffraction ( $2\theta$ ) of  $13.5^\circ$  relative to the intensity of the peak at an angle of diffraction ( $2\theta$ ) of  $11.4^\circ$  taken as 100%.

The melting point or endothermic peak of the crystals B may be, for example,  $151^\circ$  C. or more, and is preferably  $151$  to  $162^\circ$  C., as determined by differential scanning calorimetry (DSC).

The crystals B may be in any form, and may be in the form of aggregates or a pulverized powder obtained by appropriate means.

The mean particle size of the crystals B determined as a particle size at 50% ( $D_{50}$ ) in the cumulative distribution of the number of particles may be, for example,  $0.5 \mu\text{m}$  or more, and is preferably  $1$  to  $50 \mu\text{m}$ , and more preferably  $2$  to  $20 \mu\text{m}$ , but not limited thereto.

When the crystals are in the form of aggregates, the mean particle size of the crystals may be appropriately adjusted to the above range by means of, for example, pulverization.

The specific surface area of the crystals B may be, for example,  $0.2$  to  $15 \text{ m}^2/\text{g}$ , and is preferably  $0.5$  to  $10 \text{ m}^2/\text{g}$ , and more preferably  $0.9$  to  $9 \text{ m}^2/\text{g}$ , but not limited thereto.

The crystals B may have any shape, including a plate shape, a pillar shape and an acicular shape, and may preferably have a pillar shape.

The crystals of the present invention may contain both of the crystals A and the crystals B.

For such crystals, the ratio of the intensity of the peak at an angle of diffraction ( $2\theta$ ) of  $11.4^\circ$  to that of the peak at an

angle of diffraction (2 $\theta$ ) of 6.9° is, for example, 1:99 to 99:1, and may be 5:95 to 95:5, 10:90 to 90:10, 15:85 to 85:15, 20:80 to 80:20, 25:75 to 75:25, 30:70 to 70:30, 35:65 to 65:35, or 40:60 to 60:40.

The crystals may contain a higher proportion of the crystals A, and the ratio of the intensity of the peak at an angle of diffraction (2 $\theta$ ) of 11.4° to that of the peak at an angle of diffraction (2 $\theta$ ) of 6.9° may be, for example, 1:99 to 30:70, and is preferably 1:99 to 20:80, and more preferably 1:99 to 10:90.

The crystals of the present invention may contain different crystalline forms (polymorphs) of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide without departing from the scope of the invention.

### Production Method

#### Production Method for Crystals A

The present invention also includes a method for producing crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide (also simply called "compound (1)"), the method at least comprising forming a crystal nucleus in a solution containing compound (1) (this solution may also be simply called the "solution A of compound (1)") at 50° C. or more (this step may also be simply called the "nucleus formation step").

This production method typically enables efficient production of the crystals A as the crystals of compound (1) in a selective manner.

The solution A of compound (1) may be any solution that contains compound (1) dissolved in a solvent, and may be a reaction mixture containing the produced compound (1).

Examples of the solvent for dissolution of compound (1) include methyl isobutyl ketone; alcohols, such as isopropyl alcohol and butanol; hydrocarbons, such as hexane, heptane, cyclohexane, toluene, and benzene; ethers, such as tetrahydrofuran; and esters, such as ethyl acetate and butyl acetate. These solvents may be used alone or in combination of two or more of them.

The reaction mixture containing the produced compound (1) may be, for example, a reaction mixture containing compound (1) produced from 3-hydroxy-2-naphthoic acid hydrazide and methyl isobutyl ketone as precursors of compound (1).

Examples of the solvent used in the reaction include the solvents for dissolution of compound (1) as exemplified above. Methyl isobutyl ketone, which serves as a reaction precursor, is preferred as the solvent.

The concentration of compound (1) in the solution A of compound (1) is selected so that the crystals are efficiently precipitated and other advantages are obtained, and may be, for example, 20% by mass or more, and is preferably 25% by mass or more and further preferably 30% by mass or more.

The concentration of compound (1) in the solution A of compound (1) is selected so that the crystals are efficiently precipitated and other advantages are obtained, and is, for example, 40% by mass or less, preferably 38% by mass or less, and more preferably 35% by mass or less.

When the solution A of compound (1) is a reaction mixture prepared by reacting 3-hydroxy-2-naphthoic acid hydrazide with methyl isobutyl ketone, the concentration of 3-hydroxy-2-naphthoic acid hydrazide in the solution for preparing the reaction mixture is selected so that the crystals are efficiently precipitated and other advantages are

obtained, and is, for example, 15% by mass or more, preferably 19% by mass or more, and more preferably 23% by mass or more.

The concentration of 3-hydroxy-2-naphthoic acid hydrazide in the solution for preparing the reaction mixture is selected so that the crystals are efficiently precipitated and other advantages are obtained, and is, for example, 32% by mass or less, preferably 30% by mass or less, and more preferably 28% by mass or less.

The solution A of compound (1) can be prepared by, for example, heating a solution containing compound (1).

The heating temperature can be selected as appropriate for the type of solvent used, and may be any temperature at which compound (1) is dissolved in the solvent. The heating temperature is, for example, 70° C. or more, preferably 75° C. or more, and more preferably 80° C. or more, and may be 100° C. or more, 110° C. or more, or 115° C. or more.

The reaction for producing compound (1) from 3-hydroxy-2-naphthoic acid hydrazide and methyl isobutyl ketone is performed under heating, and thus the heated reaction mixture can be directly used as the solution A of compound (1). When methyl isobutyl ketone is used as the solvent for the reaction, the reaction mixture can be heated to 110° C. or more and thus is suitable as the solution A of compound (1).

The heating is performed so that compound (1) is completely dissolved, and the heating time is not specifically defined.

The nucleus formation step is typically performed to form crystal nuclei of compound (1) at 50° C. or more.

The nucleus formation step may be performed by cooling the solution A of compound (1).

The cooling rate is adjusted so that the crystals are efficiently precipitated and other advantages are obtained, and may be, for example, 0.5 to 30° C./min, and is preferably 0.5 to 20° C./min, and more preferably 0.5 to 10° C./min, but not limited thereto.

The cooling temperature may vary depending on the type of solvent used and the concentration of compound (1), and is typically 50° C. or more, preferably 55° C. or more, and further preferably 60° C. or more. For efficient formation of the crystal nuclei, the cooling temperature is preferably 90° C. or less and more preferably 85° C. or less.

The nucleus formation step is preferably performed in a manner that the solution A of compound (1) does not come into contact with oxygen, and is more preferably performed under the atmosphere of an inert gas, such as nitrogen.

After the crystal nuclei of compound (1) are formed, the crystals may be further precipitated at the same temperature, but the crystallization is preferably facilitated by rapid cooling in a crystallization step.

The cooling rate in the crystallization step is adjusted so that the crystals are efficiently precipitated and other advantages are obtained, and may be, for example, 0.5 to 30° C./min, and is preferably 0.5 to 20° C./min, and more preferably 0.5 to 10° C./min, but not limited thereto.

The cooling temperature in the crystallization step may be, for example, 30° C. or less, and is preferably 15° C. or less, and further preferably 10° C. or less, but not limited thereto.

Aging of the crystals may be performed at the cooling temperature during the crystallization step. The aging time may be, for example, 0.1 minutes to 3 hours, preferably 1 minute to 1 hour, and further preferably 5 to 30 minutes.

Aging of the crystals may be performed by, for example, stirring the solution A of compound (1) or allowing the solution A of compound (1) to stand without addition of any other components.

The crystallization may be performed by adding, as seed crystals, the crystals of compound (1) that are separately prepared in advance.

The seed crystals may be prepared by the production method of the present invention.

The amount of the seed crystals may be 10% by mass or less (for example, 0.05 to 10% by mass, 0.1 to 8% by mass, or 0.1 to 5% by mass) relative to the mass of the substrate (compound (1)), but is not limited thereto.

#### Production Method for Crystals B

The present invention also includes a method for producing crystals of compound (1), the method at least comprising precipitating crystals from a solution that contains compound (1) dissolved in a solvent containing methanol and methyl isobutyl ketone (this solution may also be simply called the "solution B of compound (1)").

This production method typically enables efficient production of the crystals B as the crystals of compound (1) in a selective manner.

The solvent may contain a solvent other than methanol or methyl isobutyl ketone.

The ratio by mass of methanol to methyl isobutyl ketone in the solvent used in the solution B of compound (1) is selected so that the crystals are efficiently precipitated and other advantages are obtained, and may be, for example, 95:5 to 10:90, and is preferably 90:10 to 30:70, and more preferably 80:20 to 50:50, but not limited thereto.

The solution B of compound (1) may be a solution that contains compound (1) dissolved in a solvent containing methanol and methyl isobutyl ketone, or may be a reaction mixture prepared by reacting 3-hydroxy-2-naphthoic acid hydrazide as a precursor of compound (1) with methyl isobutyl ketone in a solvent. This reaction mixture is preferred due to efficiency in the crystallization and other advantages.

The concentration of compound (1) in the solution B of compound (1) may be, for example, 3% by mass or more, and is preferably 5% by mass or more, but not limited thereto. The concentration of compound (1) in the solution B of compound (1) is selected so that the crystals are efficiently precipitated and other advantages are obtained, and may be, for example, 40% by mass or less, and is preferably 30% by mass or less.

When the solution B of compound (1) is a reaction mixture prepared by reacting 3-hydroxy-2-naphthoic acid hydrazide with methyl isobutyl ketone in a solvent, the concentration of 3-hydroxy-2-naphthoic acid hydrazide in the solution for preparing the reaction mixture is selected so that the crystals are efficiently precipitated and other advantages are obtained, and is, for example, 2% by mass or more (for example, 2.1% by mass or more), preferably 3% by mass or more (for example, 3.5% by mass or more), and is, for example, 30% by mass or less (for example, 28.5% by mass or less), preferably 25% by mass or less (for example, 21.5% by mass or less).

The solution B of compound (1) can be prepared by, for example, dissolving compound (1) in a solvent containing methanol and methyl isobutyl ketone under heating.

The heating temperature may be any temperature at which compound (1) is dissolved in the solvent containing methanol and methyl isobutyl ketone, and may be, but is not

limited to, for example, 40° C. or more, preferably 50° C. or more, or may be about 64.7° C., which is the boiling point of methanol as the solvent. The heating time may be, for example, about 0.1 to 5 hours, but is not limited thereto.

When the solution B of compound (1) is a solution prepared by reacting the precursors of compound (1), the reaction temperature of the solution may be, for example, 50° C. or more (for example, 60° C. or more, 70° C. or more, 80° C. or more, or 90° C. or more), and is preferably a temperature at which the solvent can be refluxed, but not limited thereto.

The reaction time may be, for example, 1 hour or more, 2 hours or more, or 3 hours or more, but is not limited thereto.

The solution B of compound (1) may be cooled to precipitate the crystals B of the present invention in the production method.

The cooling rate is adjusted so that the crystals are efficiently precipitated and other advantages are obtained, and may be, for example, 0.5 to 30° C./min, and is preferably 0.5 to 20° C./min, and more preferably 0.5 to 10° C./min, but not limited thereto.

The cooling temperature may be any temperature at which the crystals are precipitated, and may be, for example, -20° C. or more, and is preferably 0° C. or more, and may be, for example, less than 40° C., and is preferably 30° C. or less, but not limited thereto.

Aging of the crystals may be performed at the cooling temperature. The aging time is appropriately adjusted so that the crystals B of the present invention are efficiently produced, and may be, for example, 1 to 72 hours, and is preferably 1 to 48 hours, and more preferably 1 to 24 hours. The aging may be performed by, for example, stirring the solution or allowing the solution to stand without addition of any other components.

The crystallization may be performed by adding, as seed crystals, the crystals of compound (1) that are separately prepared in advance.

The seed crystals may be prepared by the production method of the present invention.

The amount of the seed crystals may be 10% by mass or less (for example, 0.05 to 10% by mass, 0.1 to 8% by mass, or 0.1 to 5% by mass) relative to the mass of the substrate (compound (1)), but is not limited thereto.

The crystals prepared by the production method as described above may be isolated by conventional solid-liquid separation (for example, filtration or centrifugation).

The crystals containing both of the crystals A and the crystals B can be produced by, for example, extending the stirring time or the aging time to a relatively longer time (for example, 5 hours or more, 10 hours or more, or 15 hours or more) after the precipitation of the crystals A in the method for producing the crystals A.

The stirring or aging temperature may be, for example, less than 50° C. (for example, 40° C. or less, or 30° C. or less), and is preferably 20° C. or less (for example, 10° C. or less), and may be, for example, -20° C. or more, and is preferably 0° C. or more, but not limited thereto.

The crystals of the present invention produced as above may be appropriately used as a rubber additive, but the application of the crystals is not limited thereto.

Accordingly, the present invention includes a rubber additive comprising the crystals of the present invention.

The rubber additive of the present invention may be appropriately used in a rubber composition, but the application of the rubber additive is not limited thereto.

Accordingly, the present invention includes a rubber composition comprising the rubber additive of the present invention.

Such a rubber composition is also defined as a composition comprising the crystals of the present invention (or a composition comprising compound (1) comprising the crystals of the present invention).

The rubber composition of the present invention comprises at least the crystals of the present invention and may further comprise crystals other than the crystals of the present invention without departing from the scope of the invention.

The amount of the crystals (the crystals A, the crystals B, or the crystals A and B) of the present invention contained in compound (1) may be, for example, 30% by mass or more, 40% by mass or more, 50% by mass or more, 60% by mass or more, 70% by mass or more, 80% by mass or more, or 90% by mass or more, but is not limited thereto.

#### Rubber Composition

The rubber composition may comprise a rubber component.

Examples of the rubber component include, but are not limited to, diene rubbers, such as polybutadiene rubber, isoprene rubber, styrene-butadiene rubber, styrene-isoprene-butadiene rubber, chloroprene rubber, and acrylonitrile-butadiene rubber; and non-diene rubbers, such as ethylene-propylene rubber and butyl rubber. These rubber components may be a natural or synthetic rubber.

These rubber components may be used alone or in combination of two or more of them.

The rubber component contained in the rubber composition of the invention is preferably a diene rubber, in particular, preferably a natural diene rubber.

The amount of the diene rubber contained as the rubber component is selected so that improved low heat build-up properties are maintained and other advantages are obtained, and is preferably 80% by mass or more and more preferably 90% by mass or more, but not limited thereto.

The rubber composition in the invention may further contain an additive other than the rubber additive of the invention.

The additional additive may be any additive commonly used in the rubber industry.

Examples of the additional additive include an antioxidant, such as a hydrazide compound other than compound (1), a phenolic antioxidant, an imidazole antioxidant, and an amine antioxidant; a filler, including inorganic fillers, such as a white filler (for example, silica, a silicate, a carbonate or a metal oxide), and organic fillers, such as carbon black; a vulcanizing agent, such as sulfur; a process oil, such as a paraffinic process oil, a naphthene process oil, and an aromatic process oil; a vulcanization accelerator, such as a thiazole vulcanization accelerator and a sulfenamide vulcanization accelerator; a vulcanization acceleration aid, such as zinc oxide and a fatty acid (for example, stearic acid); an oxidation inhibitor; an antiozonant; and a silane coupling agent.

These additional additives may be used alone or in combination of two or more of them.

The amount of the rubber additive of the invention (or the crystals of the present invention, or compound (1) comprising the crystals of the present invention) contained in the rubber composition is selected so that low heat build-up properties, good processability and other properties are obtained, and may be, for example, 0.05 to 30 parts by mass,

and is preferably 0.05 to 10 parts by mass, and more preferably 0.05 to 5 parts by mass, relative to 100 parts by mass of the rubber component, but not limited thereto.

The amount of the rubber additive of the invention (or the crystals of the present invention, or compound (1) comprising the crystals of the present invention) contained in the rubber composition is selected so that low heat build-up properties, good processability and other properties are obtained, and may be, for example, 0.02 to 18% by mass, and is preferably 0.02 to 6% by mass, and more preferably 0.02 to 3% by mass, relative to the total mass of the rubber composition, but not limited thereto.

The amount of the rubber component of the invention contained in the rubber composition of the invention is appropriately selected depending on the intended use of the rubber composition. For example, when the rubber composition is used to prevent the reduction of the elasticity of a rubber and maintain the elasticity, the amount of the rubber component is preferably 30 to 99% by mass, more preferably 40 to 80% by mass, relative to 100% by mass of the rubber composition.

#### Filler

The rubber composition of the present invention typically further contains a filler as an additive in addition to the rubber component as described above.

When the rubber composition contains a filler together with the rubber component and the crystals of the present invention, the dispersibility of the filler is improved, and the rubber composition maintains high strength, high abrasion resistance and other properties and achieves improved low heat build-up properties even in the presence of the filler.

The amount of the filler is preferably 10 to 160 parts by mass, more preferably 30 to 100 parts by mass, relative to 100 parts by mass of the rubber component, but not limited thereto. The rubber composition containing the optimized amount of the filler has further improved low heat build-up and further improved abrasion resistance. When the amount of the filler is 10 parts by mass or more, sufficient abrasion resistance is achieved. When the amount of the filler is 160 parts by mass or less, deterioration of low heat build-up properties is prevented.

The amount of the filler contained in the rubber composition is selected so that low heat build-up properties, abrasion resistance and other properties are obtained, and may be, for example, 1,000 to 16,000 parts by mass, and is preferably 2,000 to 10,000 parts by mass, and more preferably 3,000 to 7,000 parts by mass (for example, 4,000 to 6,000 parts by mass), relative to 100 parts by mass of the rubber additive of the invention (or the crystals of the present invention, or compound (1) comprising the crystals of the present invention), but not limited thereto.

The filler may be any type, and examples thereof include those exemplified above (including carbon black and silica). Carbon black is particularly preferred, which contributes to further improved low heat build-up properties, abrasion resistance and other properties.

The carbon black includes GPF, FEF, SRF, HAF, ISAF, IISAF, or SAF class carbon blacks.

The amount of the carbon black contained in the rubber composition is selected so that further improved abrasion resistance and other properties are obtained, and is preferably 10 parts by mass or more, more preferably 30 parts by mass or more, and further preferably 50 parts by mass or more, relative to 100 parts by mass of the rubber component.

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The amount of the carbon black contained in the rubber composition is selected so that further improved low heat build-up properties and processability are obtained while maintaining high abrasion resistance, and is preferably 160 parts by mass or less, more preferably 90 parts by mass or less, and further preferably 70 parts by mass or less, relative to 100 parts by mass of the rubber component.

Carbon black tends to deteriorate low heat build-up properties and processability. However, the crystals of the present invention achieve both low heat build-up and processability, and thus a relatively large amount of carbon black can be added to the rubber composition to further enhance the fracture resistance of the rubber.

The silica serving as a filler may be, for example, wet silica, dry silica, or colloidal silica, but is not limited thereto.

The inorganic filler may be, for example, an inorganic compound represented by formula (I):



wherein M is at least one selected from the group consisting of metals selected from the group consisting of Al, Mg, Ti, Ca, and Zr, oxides or hydroxides of the metals, hydrates thereof, and carbonates of the metals; n is an integer of 1 to 5; x is an integer of 0 to 10; y is an integer of 2 to 5; and z is an integer of 0 to 10.

Examples of the inorganic compound of formula (I) include aluminas ( $Al_2O_3$ ), such as  $\gamma$ -alumina and  $\alpha$ -alumina; alumina monohydrates ( $Al_2O_3 \cdot H_2O$ ), such as boehmite and diaspore; aluminum hydroxides [ $Al(OH)_3$ ], such as gibbsite and bayerite; aluminum carbonate [ $Al_2(CO_3)_3$ ]; magnesium hydroxide [ $Mg(OH)_2$ ]; magnesium oxide (MgO); magnesium carbonate ( $MgCO_3$ ); talc ( $3MgO \cdot 4SiO_2 \cdot H_2O$ ); attapulgite ( $5MgO \cdot 8SiO_2 \cdot 9H_2O$ ); titanium white ( $TiO_2$ ); titanium black ( $TiO_{2n-1}$ ); calcium oxide (CaO); calcium hydroxide [ $Ca(OH)_2$ ]; aluminum magnesium oxide ( $MgO \cdot Al_2O_3$ ); clay ( $Al_2O_3 \cdot 2SiO_2$ ); kaolin ( $Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$ ); pyrophyllite ( $Al_2O_3 \cdot 4SiO_2 \cdot H_2O$ ); bentonite ( $Al_2O_3 \cdot 4SiO_2 \cdot 2H_2O$ ); aluminum silicates, such as  $Al_2SiO_5$  and  $Al_4 \cdot 3SiO_4 \cdot 5H_2O$ ; magnesium silicates, such as  $Mg_2SiO_4$  and  $MgSiO_3$ ; calcium silicates, such as  $Ca_2SiO_4$ ; aluminum calcium silicates, such as  $Al_2O_3 \cdot CaO \cdot 2SiO_2$ ; magnesium calcium silicate ( $CaMgSiO_4$ ); calcium carbonate ( $CaCO_3$ ); zirconium oxide ( $ZrO_2$ ); zirconium hydroxide [ $ZrO(OH)_2 \cdot nH_2O$ ]; zirconium carbonate [ $Zr(CO_3)_2$ ]; various zeolites; and crystalline aluminosilicates containing an alkali metal or an alkaline earth metal.

The rubber composition may be produced by any method, for example, by kneading the components of the rubber composition. Kneading may be performed by any method, for example, by a conventional method using a kneader, such as a roller, an internal mixer, or a Banbury mixer.

The components of the rubber composition may be subjected to kneading in any order. For example, all the components may be kneaded at once, or some of the components may be previously kneaded, and then the remaining components may be kneaded.

The rubber composition of the invention can be vulcanized to give a rubber. The vulcanization may be performed by any method including a conventional method.

The rubber can be used for any applications and, in particular, may be appropriately used to produce tires (for example, a tread, a shoulder, a sidewall, a bead and a carcass of tires).

Accordingly, the present invention also includes a tire (for example, a pneumatic tire) comprising the rubber composition.

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The tire may be produced by any method including a conventional method.

## EXAMPLES

The present invention will be specifically described below with reference to Examples, but the present invention is not limited thereto.

## X-Ray Powder Diffraction Analysis

The structure of crystals was determined using an X-ray diffractometer (Rigaku Corporation, Ultima IV) with an X-ray tube with a Cu target under conditions of a tube voltage of 40 kV, a tube current of 40 mA, a sampling width of  $0.02^\circ$  and a scanning speed of  $30^\circ/\text{min}$ .

## Melting Point

The melting point of crystals was determined from the onset temperature of the endothermic peak measured with a differential scanning calorimeter (Seiko Instruments, DSC6200) while elevating the temperature from  $50^\circ \text{C}$ . to  $170^\circ \text{C}$ . at a rate of  $10^\circ \text{C}/\text{min}$  under a nitrogen stream.

Mean Particle Size ( $D_{50}$ )

The mean particle size of crystals was determined as a particle size at 50% ( $D_{50}$ ) in the cumulative distribution from sieve analysis with a laser diffraction particle size distribution analyzer HELOS & RODOS (Sympatec).

## Specific Surface Area

The specific surface area of crystals was determined by the BET method with TriStar II (Micromeritics).

## Example 1

To a reaction vessel equipped with a Dean-Stark reflux condenser and a stirrer, 138 mL of methyl isobutyl ketone and 41 g of 3-hydroxy-2-naphthoic acid hydrazide (HNH) were added so that the concentration of HNH was 27.1% by mass. The mixture was heated to about  $116^\circ \text{C}$ . and refluxed for 3 hours while removing the distilled water. The reaction mixture was cooled at a rate of  $1.7^\circ \text{C}/\text{min}$  under stirred at 250 rpm, which caused the precipitation of crystals at  $80^\circ \text{C}$ . The reaction mixture was rapidly cooled to  $5^\circ \text{C}$ . at a rate of  $1.7^\circ \text{C}/\text{min}$ . The precipitated crystals were filtered off and dried under reduced pressure to give light brown crystals (crystals A).

The crystals A produced as above had a melting point of  $157.4^\circ \text{C}$ ., a specific surface area of  $0.7 \text{ m}^2/\text{g}$ , and a mean particle size ( $D_{50}$ ) of  $11.8 \mu\text{m}$  and had a pillar shape.

The results of X-ray powder diffraction analysis of the crystals A produced in Example 1 are shown in FIG. 1 and Table 1.

TABLE 1

| $2\theta$ ( $^\circ$ ) | Relative Intensity |
|------------------------|--------------------|
| 6.9                    | 100                |
| 14.5                   | 34                 |
| 15.5                   | 9                  |
| 16.1                   | 4                  |
| 19.1                   | 5                  |
| 23.4                   | 11                 |



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TABLE 1-continued

| 2 $\theta$ (°) | Relative Intensity |
|----------------|--------------------|
| 24.1           | 18                 |
| 27.9           | 8                  |
| 28.8           | 7                  |

As is clear from FIG. 1 and Table 1, the crystals A produced in Example 1 had a peak of maximum intensity at 6.9°.

The crystals A produced in Example 1 had characteristic peaks at 6.9°, 14.5°, 15.5°, 16.1°, 19.1°, 23.4°, 24.1°, 27.9° and 28.8°.

## Example 2

To a reaction vessel equipped with a Dean-Stark reflux condenser and a stirrer, 24 ml of methyl isobutyl ketone, 126 ml of methanol and 33.0 g of 3-hydroxy-2-naphthoic acid hydrazide (HNH) were added so that the concentration of HNH was 21.7% by mass. The mixture was heated to about 64° C. and refluxed for 6 hours. The reaction mixture was cooled to 20° C. at a rate of 1.3° C./min, which caused the precipitation of crystals, and then the crystals were aged at about 20° C. for 12 hours. The precipitated crystals were filtered off and dried under reduced pressure to give light yellow crystals (crystals B).

The crystals B produced as above had a melting point of 160.1° C., a specific surface area of 0.9 m<sup>2</sup>/g, and a mean particle size (D<sub>50</sub>) of 11.7  $\mu$ m and had a pillar shape.

The results of X-ray powder diffraction analysis of the crystals B produced in Example 2 are shown in FIG. 2 and Table 2.

TABLE 2

| 2 $\theta$ (°) | Relative Intensity |
|----------------|--------------------|
| 11.2           | 63                 |
| 11.4           | 99                 |
| 13.5           | 20                 |
| 15.5           | 13                 |
| 15.8           | 26                 |
| 16.3           | 33                 |
| 17.2           | 9                  |
| 18.4           | 68                 |
| 21.6           | 16                 |
| 22.0           | 97                 |
| 25.2           | 100                |
| 26.3           | 12                 |
| 29.1           | 9                  |

As is clear from FIG. 2 and Table 2, the crystals B produced in Example 2 had a peak of maximum intensity at 25.2°.

The crystals produced in Example 2 had characteristic peaks at 11.2°, 11.4°, 13.5°, 15.5°, 15.8°, 16.3°, 17.2°, 18.4°, 21.6°, 22.0°, 26.3° and 29.1°.

## Example 3

To a reaction vessel equipped with a Dean-Stark reflux condenser and a stirrer, 138 mL of methyl isobutyl ketone and 41 g of 3-hydroxy-2-naphthoic acid hydrazide were added so that the concentration of HNH was 27.1% by mass. The mixture was heated to about 116° C. and refluxed for 3 hours while removing the distilled water. The reaction mixture was cooled at a rate of 1.7° C./min under stirring at 250 rpm, which caused the precipitation of nuclei at 80° C.

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The reaction mixture was cooled to 5° C. at a rate of 1.7° C./min. The slurry was vigorously stirred for 15 hours. After 15 hours of stirring, the crystals were filtered off and dried under reduced pressure to give light brown crystals. The crystals had the characteristic peaks of the crystals A and B. The ratio of the intensity of the peak at 11.4° to that of the peak at 6.9° for the crystals was 8:92.

## Rubber Composition and Method for Producing Rubber

The components listed in step (I) in Table 3 were mixed together in the indicated amounts (parts by mass), and the mixture was kneaded in a Banbury mixer for 3 minutes while adjusting the maximum temperature of the mixture to 150° C. by controlling the rotation rate. The mixture was conditioned until the temperature reached 80° C. or less. The components listed in step (II) in Table 3 were then added in the indicated amounts (parts by mass). The mixture was kneaded while adjusting the maximum temperature of the mixture to 110° C. or less to give an unvulcanized rubber composition. The resulting unvulcanized rubber composition was heated in a vulcanizing press at 150° C. for 30 minutes to give a rubber.

## Low Heat Build-Up (Low Heat Build-Up Index) Test

The Tan  $\delta$  value of the rubbers of Example formulations 1 and 2 was determined using a viscoelastic analyzer (Met-ravib) at a temperature of 40° C., a dynamic strain of 5% and a frequency of 15 Hz.

The Tan  $\delta$  value of the rubber of Comparative formulation 1 was also determined under the same conditions, and the resulting value was taken as 100.

The low heat build-up index of each formulation was calculated by the formula below and the results are shown in Table 3. A smaller index indicates a lower heat build-up and a smaller hysteresis loss.

$$\text{Low heat build-up index} = \left\{ \frac{\text{Tan } \delta \text{ of rubber of Example formulation 1 or 2}}{\text{Tan } \delta \text{ of rubber of Comparative formulation 1}} \right\} \times 100$$

## Mooney Viscosity (Processability Index) Test

The Mooney viscosity of the unvulcanized rubber compositions of Example formulations 1 and 2 was determined using a Mooney viscometer at 100° C. The Mooney viscosity of the unvulcanized rubber composition of Comparative formulation 1 was also determined in the same manner as above, and the resulting value was taken as 100. The processability index was calculated by the formula below and the results are shown in Table 3. A smaller index indicates better processability and higher productivity of an unvulcanized rubber composition.

$$\text{Processability index} = \left\{ \frac{\text{Mooney viscosity of unvulcanized rubber composition of Example formulation 1 or 2}}{\text{Mooney viscosity of unvulcanized rubber composition of Comparative formulation 1}} \right\} \times 100$$

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TABLE 3

|                         |      |                            | Example<br>formulation<br>1 | Example<br>formulation<br>2 | Compara-<br>tive<br>formulation<br>1 |
|-------------------------|------|----------------------------|-----------------------------|-----------------------------|--------------------------------------|
| Comp-                   | Step | NR (TSR20)* <sup>1</sup>   | 100                         | 100                         | 100                                  |
| osition                 | (I)  | CB (N234)* <sup>2</sup>    | 45                          | 45                          | 45                                   |
|                         |      | Antioxidant* <sup>3</sup>  | 1                           | 1                           | 1                                    |
|                         |      | Wax* <sup>4</sup>          | 2                           | 2                           | 2                                    |
|                         |      | Stearic acid* <sup>5</sup> | 2                           | 2                           | 2                                    |
|                         |      | Crystals A* <sup>6</sup>   | 1                           |                             |                                      |
|                         |      | Crystals B* <sup>7</sup>   |                             | 1                           |                                      |
|                         | Step | Vulcanization              | 1.2                         | 1.2                         | 1.2                                  |
|                         | (II) | accelerator                |                             |                             |                                      |
|                         |      | (TBBS)* <sup>8</sup>       |                             |                             |                                      |
|                         |      | Sulfur* <sup>9</sup>       | 2                           | 2                           | 2                                    |
|                         |      | Zinc oxide* <sup>10</sup>  | 3                           | 3                           | 3                                    |
| Low heat build-up index |      |                            | 87                          | 90                          | 100                                  |
| Processability index    |      |                            | 115                         | 113                         | 100                                  |

\*<sup>1</sup>NR: Natural rubber, TSR20.\*<sup>2</sup>CB (carbon black): N234 (Cabot Corporation)\*<sup>3</sup>Antioxidant: N-phenyl-N'-(1,3-dimethylbutyl)-p-phenylenediamine (Kemai Chemical Co., Ltd.)\*<sup>4</sup>Wax: Antilux 111 (Rhein Chemie Rheinau GmbH)\*<sup>5</sup>Stearic acid (Sichuan Tianyu Grease Chemistry Co., Ltd.)\*<sup>6</sup>Crystals A of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide produced in Example 1.\*<sup>7</sup>Crystals B of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide produced in Example 2.\*<sup>8</sup>Vulcanization accelerator (TBBS): N-(tert-butyl)-2-benzothiazole sulfenamide (SANCER NS-G, Ouchi Shinko Chemical Industrial Co., Ltd.)\*<sup>9</sup>Sulfur (Shanghai Jinghai Chemical Co., Ltd.)\*<sup>10</sup>Zinc oxide (Dalian Zinc Oxide Co., Ltd.)

As apparent from the results in the table, the rubber of Example formulation 1 containing the crystals A and the rubber of Example formulation 2 containing the crystals B had improved low heat build-up as compared with that of the rubber of Comparative formulation 1 not containing the crystals of compound (1).

The rubber of Example formulation 1 containing the crystals A had improved low heat build-up as compared with that of the rubber of Example formulation 2 containing the crystals B. The unvulcanized rubber of Example formulation 2 containing the crystals B had better processability as compared with that of the unvulcanized rubber of Example formulation 1 containing the crystals A.

## INDUSTRIAL APPLICABILITY

The crystals of the present invention can be used as a rubber additive capable of imparting low heat build-up properties and other properties to a rubber component without severely deteriorating processability and are thus very useful in industries.

The invention claimed is:

1. Crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising a peak of maximum intensity at an angle of diffraction (2θ) of 6.9±0.2°.

2. The crystals according to claim 1, wherein the crystals have a melting point of 151° C. or more.

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3. Crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising characteristic peaks at least at angles of diffraction (2θ) of 6.9±0.2°, 14.5±0.2°, 15.5±0.2°, 19.1±0.2°, 23.4±0.2°, 24.1±0.2°, 27.9±0.2° and 28.8±0.2°.

4. Crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising a peak of maximum intensity at an angle of diffraction (2θ) of 11.4±0.2° and/or 25.2±0.2°.

5. The crystals according to claim 4, wherein the crystals have a melting point of 151° C. or more.

6. Crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising characteristic peaks at least at angles of diffraction (2θ) of 11.2±0.2°, 11.4±0.2°, 13.5±0.2°, 15.5±0.2°, 15.8±0.2°, 16.3±0.2°, 18.4±0.2°, 21.6±0.2°, 22.0±0.2°, 25.2±0.2° and 26.3±0.2°.

7. Crystals comprising crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising a peak of maximum intensity at an angle of diffraction (2θ) of 6.9±0.2° and crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising a peak of maximum intensity at an angle of diffraction (2θ) of 11.4±0.2° and/or 25.2±0.2°.

8. The crystals according to claim 7, wherein a ratio of the intensity of the peak at an angle of diffraction (2θ) of 11.4±0.2° to that of the peak at an angle of diffraction (2θ) of 6.9±0.2° is 1:99 to 10:90.

9. Crystals comprising crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising characteristic peaks at least at angles of diffraction (2θ) of 6.9±0.2°, 14.5±0.2°, 15.5±0.2°, 19.1±0.2°, 23.4±0.2°, 24.1±0.2°, 27.9±0.2° and 28.8±0.2° and crystals of 3-hydroxy-N'-(1,3-dimethylbutylidene)-2-naphthoic acid hydrazide having an X-ray powder diffraction pattern comprising characteristic peaks at least at angles of diffraction (2θ) of 11.2±0.2°, 11.4±0.2°, 13.5±0.2°, 15.5±0.2°, 15.8±0.2°, 16.3±0.2°, 18.4±0.2°, 21.6±0.2°, 22.0±0.2°, 25.2±0.2° and 26.3±0.2°.

10. The crystals according to claim 9, wherein a ratio of the intensity of the peak at an angle of diffraction (2θ) of 11.4±0.2° to that of the peak at an angle of diffraction (2θ) of 6.9±0.2° is 1:99 to 10:90.

11. A rubber additive comprising the crystals according to claim 1.

12. A rubber additive comprising the crystals according to claim 4.

13. A rubber composition comprising the rubber additive according to claim 11.

14. A rubber composition comprising the rubber additive according to claim 12.

15. A tire comprising the rubber composition according to claim 13.

16. A tire comprising the rubber composition according to claim 14.

\* \* \* \* \*